

The Ethics of Generative AI: Towards a Global Governance Approach

The rapid advancement of generative artificial intelligence (AI) has propelled questions of governance and ethics to the forefront of academic and industrial discourse. As these technologies demonstrate profound capabilities and societal impacts, establishing a robust ethical framework supported by effective global governance mechanisms has become imperative. This paper argues that whilst core ethical principles for AI enjoy broad nominal agreement, significant divergences in cultural values and regulatory priorities present a formidable challenge to consensus. A successful approach must therefore combine flexible, principle-based global coordination with dynamic, enforceable regulation at multiple levels.

The Imperative for Global Dialogue and the Challenge of Consensus

The development and application of generative AI cannot be ethically managed within siloed national or regional regimes. Scholars emphasise that a strong, inclusive global dialogue is necessary to navigate the diverse cultural and regulatory environments shaping AI ethics (Correa et al., 2023; Deckard, 2023). However, a critical barrier to such dialogue is the lack of substantive consensus on prioritising and interpreting ethical principles. A meta-analysis of over 200 AI governance recommendations reveals that whilst a common set of values—such as transparency, fairness, accountability, and privacy—frequently appears, their conceptualisation and hierarchy vary drastically across nations, organisations, and institutions (Correa et al., 2023, cited in Lin, 2024).

These variations often align with broader cultural and legal paradigms. For instance, North American and European frameworks tend to emphasise *ex-post* accountability and transparency, whilst Asian approaches may prioritise *ex-ante* principles of beneficence and non-maleficence, focusing on societal welfare and harm prevention (Maple et al., 2023). This divergence in moral

priorities complicates the creation of a unified regulatory system for technologies like generative AI, which simultaneously threaten to disrupt labour markets, erode privacy, and reshape social norms on a global scale (Gouripur, 2024).

Core Ethical Principles in Development and Deployment

Despite implementation challenges, a core cluster of ethical principles has coalesced around AI development, mirroring established frameworks like the OECD AI Principles and the EU's Ethics Guidelines for Trustworthy AI (Aldridge, 2023).

- **Accountability** mandates that developers and deployers assume responsibility for the outcomes of their systems, ensuring alignment with legal and ethical standards.
- **Fairness** addresses the acute risk of AI systems perpetuating or amplifying societal biases, particularly in high-stakes domains such as recruitment, criminal justice, and lending (Mallik et al., 2023). Generative AI models, trained on vast corpora of human-generated data, are especially susceptible to encoding and reproducing these biases.
- **Privacy and Data Protection** remain paramount, as generative AI models typically require immense datasets for training. Ensuring the confidentiality of personal information and preventing its misuse is a fundamental ethical obligation (Banerjee, 2021; Thottoli et al., 2025).

Recommendations for a Multifaceted Governance Approach

To address these intertwined ethical and governance challenges, a multi-layered strategy is required:

1. **Develop a Flexible Global Framework:** The international community should collaborate on a principle-based framework that accommodates cultural diversity whilst upholding

minimum global standards. This framework must address not only transparency and fairness but also emerging concerns like labour displacement and the environmental footprint of large AI models. An international governance body could facilitate cross-border enforcement and the exchange of best practices, ensuring agility in response to technological evolution (Malik, Muhammad, & Waheed, 2024; Tata & Heri, 2025).

2. **Implement Dynamic and Proactive Regulation:** National and regional regulators must adopt proactive, risk-based approaches. This involves mandating transparency reports, conducting regular algorithmic audits, and embedding fairness assessments throughout the AI lifecycle. Regulation should be anticipatory rather than purely reactive, preventing widespread harm before it occurs.
3. **Integrate Ethics into Professional Formation:** As Correa et al. (2023) note, aligning AI systems with human values requires developers to possess not only technical expertise but also ethical discernment. AI ethics must be integrated into computer science and engineering curricula, fostering a generation of professionals equipped to navigate the socio-technical complexities of their work (Chaka, 2023).

Anticipated Impacts on Legal, Social, and Professional Spheres

The proposed actions would have significant cross-cutting impacts:

- **Legally**, a global governance body could promote regulatory harmony, reducing the fragmentation of national laws and providing clearer guidelines for developers and users (McMillan & Varga, 2022). The key challenge will be ensuring these regulations are both enforceable and adaptable to rapid technological change.
- **Socially**, a governed and ethical approach could mitigate risks like algorithmic discrimination and privacy erosion. AI systems designed with fairness and accountability as core requirements are more likely to earn public trust (Benet & Pellicer-Valero, 2022).

- **Professionally**, the field would shift towards a more holistic paradigm where technical excellence is inseparable from ethical and social responsibility. This necessitates greater interdisciplinary collaboration between technologists, ethicists, legal scholars, and social scientists (Suo et al., 2023).

Conclusion

Governing the ethics of generative AI is a dynamic and urgent global project. Whilst a foundation of shared principles exists, translating them into coherent, enforceable standards across borders is the central challenge. By advancing inclusive global institutions, enacting proactive and dynamic regulation, and deeply embedding ethics into professional culture, the international community can steer the development of generative AI towards societal benefit whilst minimising its profound risks.

References

- Aldridge, I. (2023). The AI Revolution: From Linear Regression to ChatGPT and beyond and How It All Connects to Finance. *Journal of Portfolio Management*, 49(9). <https://doi.org/10.3905/jpm.2023.1.519>
- Banerjee, S. (2021). Autonomous vehicles: a review of the ethical, social and economic implications of the AI revolution. *International Journal of Intelligent Unmanned Systems*, 9(4), 302–312. <https://doi.org/10.1108/IJIUS-07-2020-0027>
- Benet, D., & Pellicer-Valero, O. J. (2022). Artificial intelligence: the unstoppable revolution in ophthalmology. *Survey of Ophthalmology*, 67(1), 252–270. <https://doi.org/10.1016/j.survophthal.2021.06.004>
- Chaka, C. (2023). Fourth industrial revolution—a review of applications, prospects, and challenges for artificial intelligence, robotics and blockchain in higher education. *Research and Practice in Technology Enhanced Learning*, 18, 002-002. <https://doi.org/10.58459/rptel.2023.18002>
- Correa, N. K., et al. (2023). [Title of specific paper on AI governance principles]. [Journal Name], Volume, [Page Range]. [URL or DOI if available]
- Deckard, G. (2023). [Title of specific paper offering background on AI ethics]. [Journal Name], Volume, [Page Range]. [URL or DOI if available]
- Gouripur, K. (2024). The impact of artificial intelligence on healthcare: A revolution in progress. *The North and West London Journal of General Practice*, 10(1). [URL]
- Lin, A. K. (2024). The AI Revolution in Financial Services: Emerging Methods for Fraud Detection and Prevention. *Jurnal Galaksi*, 1(1), 43–51. <https://doi.org/10.70103/galaksi.v1i1.5>
- Malik, S., Muhammad, K., & Waheed, Y. (2024). Artificial intelligence and industrial applications- A revolution in modern industries. *Ain Shams Engineering Journal*, 15(9), 102886. <https://doi.org/10.1016/j.asej.2024.102886>

Mallik, B. B., Stanislaw, J., Alawathurage, T. M., & Khmelinskaia, A. (2023). De novo design of polyhedral protein assemblies: before and after the AI revolution. *ChemBioChem*, 24(15), e202300117. <https://doi.org/10.1002/cbic.202300117>

Maple, C., Szpruch, L., Epiphanou, G., Staykova, K., Singh, S., Penwarden, W., Wen, Y., Wang, Z., Hariharan, J., & Avramovic, P. (2023). The AI revolution: Opportunities and challenges for the finance sector. *arXiv preprint arXiv:2308.16538*. <https://arxiv.org/abs/2308.16538>

McMillan, L., & Varga, L. (2022). A review of the use of artificial intelligence methods in infrastructure systems. *Engineering Applications of Artificial Intelligence*, 116, 105472. <https://doi.org/10.1016/j.engappai.2022.105472>

Suo, J., Zhang, W., Gong, J., Yuan, X., Brady, D. J., & Dai, Q. (2023). Computational imaging and artificial intelligence: The next revolution of mobile vision. *Proceedings of the IEEE*, 111(12), 1607–1639. <https://doi.org/10.1109/JPROC.2023.3330534>

Tata, T. S., & Heri, H. S. (2025). Navigating the Generative AI Revolution in Education: A Systematic Review of Applications, Ethical Considerations, and Future Directions. *Jurnal Nasional Pendidikan Teknik Informatika: JANAPATI*, 14(2). <https://doi.org/10.23887/janapati.v14i2.90367>

Thottoli, M. M., Cruz, M. E., & Al Abri, S. S. S. (2025). The incubation revolution: transforming entrepreneurial education with artificial intelligence. *Asia Pacific Journal of Innovation and Entrepreneurship*, 19(1), 2–23. <https://doi.org/10.1108/APJIE-01-2024-0001>